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NEW-AGE GLOBAL UNIVERSITY FOR LIBERAL EDUCATION

Semester: II

Category: Core

Operating System

Course Code: CS1103

(Theory and Lab)

Course Outline

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Unit – I	04 Hrs
Introduction to Operating Systems: Operating Systems –Definition Types- Functions -Abstract view of OS- System Structures – System Calls- Virtual Machines.	
Unit – II	08 Hrs
Process Management: Process Concepts, Inter-process communication –Threads –Multithreading, Process Scheduling, First-Come, First-Served (FCFS), Shortest Job First, Round Robin (RR), Priority Scheduling.	
Unit – III	06 Hrs
Synchronization and Deadlocks: Synchronization –Semaphores –Monitors, Hardware Synchronization – Deadlocks –Methods for Handling Deadlocks.	
Unit – IV	12 Hrs
Memory Management: Memory Management Strategies –Contiguous and Non-Contiguous allocation –Paging –Virtual memory Management –File System Design and Implementation – Mass Storage Structure –Disk Scheduling and Management, FCFS, SSTF, and SCAN -I/O Systems- Overview of System Protection and Security.	

Contents

- When is a process created?
- Activity: Display of System daemons
- Process Creation
- When do processes end?
- Process Termination
- Hierarchy of process

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Created by: Dr. Sangeeta Bhanja Chaudhuri

When is a process created?

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Processes can be created in two ways

- System initialization: when the OS starts up, one or more processes are created.
- Execution of a process creation system call: something explicitly asks for a new process

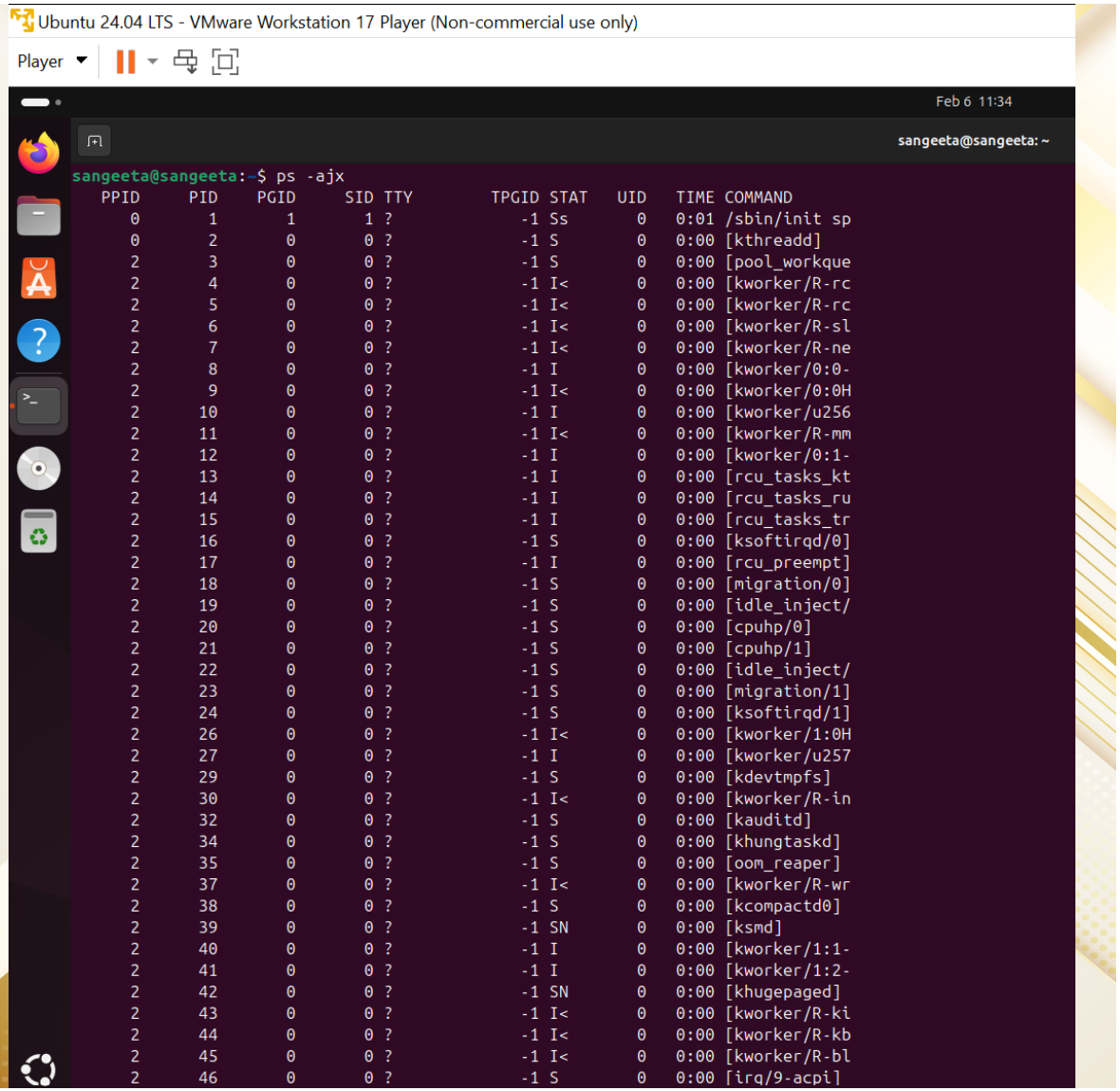
System calls can come from

- User request to create a new process (system call executed from user shell)
- Already running processes
  - User programs
  - System daemons // background program of a computer's operating system, performing tasks without the user's direct interaction.

Courtesy: CS 1550, cs.pitt.edu (originally modified by Ethan L. Miller and Scott A. Brandt)

Activity: Display of System daemons

Note: PID - Process ID
PPID - Parent Process ID
PGID - Process Group ID
SID - Session ID
UID - User ID
EUID - Effective User ID
TTY- Teletypewriter (name of terminal)
Target Portal Group (TPG) ID



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Process Creation

- Parent process creates children processes, which, in turn create other processes, forming a tree of processes
- Generally, process identified and managed via a process identifier (pid)
- Resource sharing options
  - Parent and children share all resources
  - Children share subset of parent's resources
  - Parent and child share no resources
- Execution options
  - Parent and children execute concurrently
  - Parent waits until children terminate

Courtesy: CS/COE 1550 – Operating Systems – Sherif Khattab

Process Creation

- Address space
  - Child duplicate of parent
  - Child has a program loaded into it
- UNIX examples
  - **`fork()`** system call creates new process
  - **`exec()`** system call used after a **`fork()`** to replace the process' memory space with a new program

Courtesy: CS/COE 1550 – Operating Systems – Sherif Khattab

When do processes end?

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Conditions that terminate processes can be

- Voluntary
- Involuntary

Voluntary

- Normal exit
- Error exit

Involuntary

- Fatal error (only sort of involuntary)
- Killed by another process

Courtesy: CS 1550, cs.pitt.edu (originally modified by Ethan L. Miller and Scott A. Brandt)

When do processes end?

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- Some operating systems do not allow a child to exist if its parent has terminated. If a process terminates, then all its children must also be terminated.
  - **cascading termination.** All children, grandchildren, etc. are terminated.
  - The termination is initiated by the operating system.
- The parent process may wait for termination of a child process by using the **`wait()`** system call . The call returns 'status information' and the pid of the terminated process

```
pid = wait(&status);
```

- If no parent waiting (did not invoke **`wait()`**) process is a **zombie**
- If parent terminated without invoking **`wait`** , process is an **orphan**

Courtesy: CS 1550, cs.pitt.edu (originally modified by Ethan L. Miller and Scott A. Brandt)

Process Termination

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- Process executes last statement and then asks the operating system to delete it using the **`exit()`** system call.
  - Returns status data from child to parent (via **`wait()`**)
  - Process' resources are deallocated by operating system
- Parent may terminate the execution of children processes using the **`abort()`** system call. Some reasons for doing so:
  - Child has exceeded allocated resources
  - Task assigned to child is no longer required
  - The parent is exiting and the operating systems does not allow a child to continue if its parent terminates

Courtesy: CS/COE 1550 – Operating Systems – Sherif Khattab

Hierarchies of Process

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Parent creates a child process

- Child processes can create their own children

Forms a hierarchy

- UNIX calls this a “process group”
- If a process exits, its children are “inherited” by the exiting process’s parent

Windows has no concept of process hierarchy

- All processes are created equal

Activity

Execute a program that calls a daemonize function to daemonize itself as a daemon.

Refer to: https://notes.shichao.io/apue/ch13/

Created by: Dr. Sangeeta Bhanja Chaudhuri

References

- Silberschatz, Galvin, Gagne, Operating System Concepts, John Wiley and Sons, 10th edition, 2023, ISBN: 935746056X
- Modern Operating Systems by Tanenbaum and Bos , 4th edition, Pearson Publisher, ISBN-13: 978-0-13-359162-0
- Courtesy: Prof. Mouli Sankaran, RV University.

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